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By Amit Katiyar (MCA-JNU)



MATHEMATICS

- The point of intersection of circle $x^2 + y^2 + 10x - 12y + 51 = 0$ and the line $3y + x = 3$ is
(a) $(-6, 3)$ (b) $(3, -6)$
(c) $(6, -3)$ (d) $(-3, 6)$
- The number of solution of the equation $\sin x + \sin 5x = \sin 3x$ lying in the interval $[0, \pi]$ is
(a) 4 (b) 6 (c) 5 (d) 2
- In an acute angled $\triangle ABC$ the least value of $\sec A + \sec B + \sec C$ is
(a) 6 (b) 8 (c) 3 (d) 2
- Let $P = \{\theta: \sin \theta - \cos \theta = \sqrt{2} \cos \theta\}$ and $Q = \{\theta: \sin \theta + \cos \theta = \sqrt{2} \sin \theta\}$ be two sets. Then
(a) $P \subset Q$ and $Q - P \neq \phi$
(b) $P \not\subset Q$
(c) $Q \not\subset P$
(d) $P = Q$
- If $\frac{\tan x}{2} = \frac{\tan y}{3} = \frac{\tan z}{5}$ and $x + y + z$, then the value of $\tan^2 x + \tan^2 y + \tan^2 z$ is
(a) $\frac{38}{3}$ (b) 38
(c) 114 (d) None of these
- The circles whose equations are $x^2 + y^2 + c^2 = 2ax$ and $x^2 + y^2 + c^2 - 2by = 0$ will touch another externally if
(a) $\frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{a^2}$ (b) $\frac{1}{c^2} + \frac{1}{a^2} = \frac{1}{b^2}$
(c) $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{c^2}$ (d) None of these
- The locus of the orthocenter of the triangle formed by the lines $(1 + p)x - py + p(1 + p) = 0$, $(1 + q)(x - q) + q(1 + q) = 0$ and $y = 0$ where $p \neq q$ is
(a) a hyperbola (b) a parabola
(c) an ellipse (d) a straight line
- Equation of a common tangent with positive slope to the circle $x^2 + y^2 - 8x = 0$ as well as to the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ is
(a) $2x - \sqrt{5}y - 20 = 0$ (b) $2x - \sqrt{5}y + 4 = 0$
(c) $3x - 4y + 8 = 0$ (d) $4x - 3y + 4 = 0$
- The area enclosed between the curves $y^2 = x$ and $y = |x|$ is
(a) $\frac{2}{3}$ sq. unit (b) 1 sq. unit
(c) $\frac{1}{6}$ sq. unit (d) $\frac{1}{3}$ sq. unit
- Equation of the perpendicular to $x - 2y = 1$ And passing through $(1, 1)$ is
(a) $x + 2y = 3$ (b) $x + y = 2$
(c) $y = 2x + 3$ (d) $y = -2x + 3$
- If $A = \begin{bmatrix} 0 & 5 \\ 0 & 0 \end{bmatrix}$ & $f(x) = 1 + x + x^2 + \dots + x^{16}$, then $f(A) =$
(a) 0 (b) $\begin{bmatrix} 1 & 5 \\ 0 & 1 \end{bmatrix}$
(c) $\begin{bmatrix} 1 & 5 \\ 0 & 0 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 5 \\ 0 & 1 \end{bmatrix}$

- 9 balls are to be placed in 9 boxes and 5 of the balls cannot fit into 3 small boxes. The number of ways of arranging one ball in each of the boxes is
(a) 18720 (b) 18270 (c) 17280 (d) 12780
- Which of the following functions is inverse of itself?
(a) $f(x) = \frac{1-x}{1+x}$ (b) $f(x) = 3^{\log x}$
(c) $f(x) = 3^{x(x+1)}$ (d) None of these
- A student council has 10 members. From this one President, one vice-president, one Secretary, one Joint-Secretary and two Executive Committee members have to be elected. In how many ways this can be done?
(a) 151200 (b) 75600
(c) 37800 (d) 18900
- In a survey where 100 students reported which subjects they like, 32 students in total liked Mathematics, 38 students liked business and 30 students liked literature. Moreover 7 students liked both Mathematics and Literature, 10 students like both Mathematics and Business, 8 students liked both Business and Literature, 5 students like all three subjects. Then the number of people who liked exactly one subject is
(a) 60 (b) 65 (c) 70 (d) 78
- The number of natural numbers which are smaller than 2×10^8 and which contain only the digits 1 and 2 is
(a) 786 (b) 666 (c) 766 (d) 1066
- If $\lim_{x \rightarrow \infty} \left(1 + \frac{a}{x} + \frac{b}{x^2}\right)^{2x} = e^2$, then the value of a and b are
(a) $a \in R, b = 2$ (b) $a = 1, b \in R$
(c) $a \in R, b \in R$ (d) $a = 1, b = 2$
- If $f(x) = \sin^5 x + \sin^3 x$ and $g(x) = \cos^5 x + \sin^3 x$, then the value of $\int_0^{\frac{\pi}{2}} [f(x) + f(-x)][g(x) + g(-x)] dx$ is
(a) 0 (b) > 1
(c) 0 and 1 (d) less than 0
- $\frac{d^2 x}{dy^2}$ equals
(a) $\left(\frac{d^2 y}{dx^2}\right)^{-1}$ (b) $\left(\frac{d^2 y}{dx^2}\right)^{-1} \left(\frac{dy}{dx}\right)^{-3}$
(c) $\left(\frac{d^2 y}{dx^2}\right) \left(\frac{dy}{dx}\right)^{-2}$ (d) $-\left(\frac{d^2 y}{dx^2}\right) \left(\frac{dy}{dx}\right)^{-3}$
- Differential coefficient of $\log_{10} x$ w.r. to $\log_x 10$ is
(a) $-\frac{(\log x)^2}{(\log 10)^2}$ (b) $\frac{(\log_{10} x)^2}{(\log 10)^2}$
(c) $\frac{(\log x 10)^2}{(\log 10)^2}$ (d) $-\frac{\log 10^2}{(\log x)^2}$
- $f(x) = x + |x|$ is continuous for
(a) $x \in (-\infty, \infty)$ (b) $x \in (-\infty, \infty) - \{0\}$
(c) Only $x > 0$ (d) No value of x
- If a, b and c are unit vectors, then $|a - b|^2 + |b - c|^2 + |c - a|^2$ does not exceed
(a) 4 (b) 9 (c) 8 (d) 6





23. The vector $\vec{a} = \alpha\hat{i} + 2\hat{j} + \beta\hat{k}$ lies in the plane of the vectors $\vec{b} = \hat{i} + \hat{j}$ and $\vec{c} = \hat{j} + \hat{k}$ and bisects the angle between $\vec{b}$ and $\vec{c}$. Then, which one of the following gives possible values of α and β ?
- (a) $\alpha = 2, \beta = 2$ (b) $\alpha = 1, \beta = 2$
(c) $\alpha = 2, \beta = 1$ (d) $\alpha = 1, \beta = 1$
24. Forces $4\hat{i} - 3\hat{j} + 7\hat{k}$ and $-2\hat{i} + 2\hat{j} - 8\hat{k}$ are acting on a particle and displace it from the point (5,7,1) to (2,5,-6), then the work done by the force is
- (a) 3 (b) 9 (c) 15 (d) 7
25. A bird is flying in a straight line with velocity vector $10\hat{i} + 6\hat{j} + \hat{k}$, measured in km/h. If starting point is (1,2,3), how much time does it take to reach a point in space that is 13m high from ground?
- (a) 600 seconds (b) 360 seconds
(c) 36 seconds (d) 60 seconds
26. If A and B are two events and $P(A \cup B) = \frac{5}{6}, P(A \cap B) = \frac{1}{3}, P(B) = \frac{1}{2}$, then A and B are two events which are
- (a) dependent (b) Independent
(c) Mutually exclusive (d) Equally likely
27. If $a_1, a_2, a_3, \dots, a_n$ are positive real numbers whose product is a fixed number C, then the minimum value of $a_1 + a_2 + \dots + a_n$ is
- (a) $n(2C)^{1/n}$ (b) $(n+1)C^{1/n}$
(c) $2nC^{1/n}$ (d) $(n+1)(2C)^{1/n}$
28. If a, b, c are the roots of the equation $x^3 - 3x^2 + 3x + 7 = 0$, then the value of
- $$\begin{vmatrix} 2bc - a^2 & c^2 & b^2 \\ c^2 & 2ac - b^2 & a^2 \\ b^2 & a^2 & 2ab - c^2 \end{vmatrix}$$
- is
- (a) 9 (b) 27 (c) 81 (d) 0
29. The coefficient of x^n in the expansion of $(1 - 2x + 3x^2 - 4x^3 + \dots \text{to } \infty)^{-n}$ is
- (a) $\frac{(2n)!}{n!(n-1)!}$ (b) $\frac{(2n)!}{[(n-1)!]^2}$
(c) $\frac{(2n)!}{(n!)^2}$ (d) None of these
30. Let α, β be the roots of the equation $x^2 - px + r = 0$ and $\frac{\alpha}{2}, 2\beta$ be the roots of the equation $x^2 - qx + r = 0$. Then, the value of r is
- (a) $\frac{2}{9}(p-q)(2q-p)$ (b) $\frac{2}{9}(q-p)(2q-p)$
(c) $\frac{2}{9}(p-2p)(2q-p)$ (d) $\frac{2}{9}(2p-q)(2q-p)$
31. The value of $\cot\left(\operatorname{cosec}^{-1}\frac{5}{3} + \tan^{-1}\frac{2}{3}\right)$ is
- (a) 6/17 (b) 3/17
(c) 4/17 (d) 5/17
32. If $\sin\theta = 3\sin(\theta + 2\alpha)$, then the value of $\tan(\theta + \alpha) + 2\tan\alpha$ is
- (a) 3 (b) 2 (c) -1 (d) 0
33. In a ΔABC , $\angle A = 90^\circ$ and D is mid-point of AC. The value of $BC^2 - BD^2$ is equal to
- (a) AD^2 (b) $2AD^2$ (c) $3AD^2$ (d) $4AD^2$
34. Through any point (x, y) of a curve which passes through the origin, lines are drawn parallel to the coordinate axes. The curve given that divides the rectangle formed by the two lines and the two lines and the axis into two areas, one of the which is twice the other, represents a family of
- (a) circles (b) parabolas
(c) hyperbolas (d) straight lines
35. A line through (4, 2) meets coordinate axes at P and Q. Then the locus of the circumcentre of ΔOPQ is
- (a) $\frac{1}{x} + \frac{1}{y} = 2$ (b) $\frac{2}{x} + \frac{1}{y} = 1$
(c) $\frac{1}{x} + \frac{2}{y} = 1$ (d) $\frac{1}{x} + \frac{1}{y} = \frac{1}{2}$
36. $6 + \log_{\frac{1}{4}} \frac{1}{\sqrt{2}} \left[\sqrt{1 - \frac{1}{\sqrt{2}}} \sqrt{1 - \frac{1}{\sqrt{2}}} \sqrt{1 - \frac{1}{\sqrt{2}}} \dots \right] =$
- (a) 6 (b) $\frac{13}{2}$ (c) 4 (d) $\frac{25}{4}$
37. The function $f(x) = \log(x + \sqrt{x^2 + 1})$ is
- (a) an even function
(b) an odd function
(c) a periodic function
(d) Neither an even nor an odd function
38. Two persons A and B agree to meet on 20th April 2018 between 6 P.M. and 7 P.M. with the understanding that they will wait no longer than 20 minutes for the other. What is the probability that they meet?
- (a) 5/9 (b) 7/9
(c) 2/9 (d) 4/9
39. Three numbers a, b and c are chosen at random (without replacement) from among the numbers 1, 2, 3, ..., 99. the probability that $a^3 + b^3 + c^3 - 3abc$ is divisible by 3 is
- (a) $\frac{3 \cdot {}^{33}C_3 + ({}^{33}C_1)^3}{99C_3}$ (b) $\frac{3 \cdot {}^{33}C_3 - ({}^{33}C_1)^3}{99C_3}$
(c) $\frac{2 \cdot {}^{33}C_3 + ({}^{33}C_1)^3}{99C_3}$ (d) $\frac{2 \cdot {}^{33}C_3 - ({}^{33}C_1)^3}{99C_3}$
40. A and B play a game where each is asked to select a number from 1 to 25. If the two numbers match, both of them win a prize. The probability that they will not win a prize in a single trial is
- (a) $\frac{1}{25}$ (b) $\frac{24}{25}$
(c) $\frac{2}{25}$ (d) None
41. The quadratic equation whose roots are $\sin^2 18^\circ$ and $\cos^2 36^\circ$ is
- (a) $16x^2 - 12x + 1 = 0$
(b) $16x^2 + 12x + 1 = 0$
(c) $16x^2 - 12x - 1 = 0$
(d) $16x^2 + 10x + 1 = 0$



42. Sum to infinity of a geometric progression is twice the sum of the first two terms. Then, what are the possible values of the common ratio?

- (a) $\pm \frac{1}{\sqrt{2}}$ (b) $\pm \frac{1}{2}$
(c) $\pm \frac{1}{\sqrt{3}}$ (d) $\pm \frac{1}{3}$

43. Suppose that m and n are fixed numbers such that the m^{th} term a_m is equal to n and the n^{th} term a_n is equal to m ($m \neq n$), then the $(m+n)^{\text{th}}$ term is

- (a) $\frac{(m+n)}{mn}$ (b) $mn/(m+n)$
(c) $(m+n)/n$ (d) $(m+n)/m$

44. If A is an invertible skew symmetric matrix, then A^{-1} is a

- (a) symmetric matrix (b) skew-symmetric matrix
(c) zero matrix (d) identity matrix

45. If the mean of squares of first n natural numbers be 11, then n is equal to

- (a) $-\frac{13}{2}$ (b) 11 (c) 5 (d) 4

46. The set of points, where $f(x) = \frac{x}{1+|x|}$ is differentiable is

- (a) $(-\infty, -1) \cup (-1, \infty)$ (b) $(-\infty, \infty)$
(c) $(0, \infty)$ (d) $(-\infty, 0) \cup (0, \infty)$

47. $\int_0^\pi x f(\sin x) dx$ is equal to

- (a) $\pi \int_0^\pi f(\sin x) dx$ (b) $\frac{\pi}{2} \int_0^\pi f(\sin x) dx$
(c) $\pi \int_0^{\pi/2} f(\cos x) dx$ (d) $\pi \int_0^\pi x f(\sin x) dx$

48. Let $f: R \rightarrow R$ be defined by $f(x) = \begin{cases} x+2 & \text{if } x < 0 \\ |x-2| & \text{if } x \geq 0 \end{cases}$ Find $\int_{-2}^3 f(x) dx$.

- (a) 0.5 (b) 2.5 (c) 4.5 (d) 6.5

49. Slope of two lines $6x^2 - 2xy - 2y^2 = 0$ differ by

- (a) $\frac{5}{2}$ (b) $\frac{7}{2}$ (c) $\frac{7}{5}$ (d) -1

50. If the radius of a circle changes at the rate of $-\frac{2}{\pi}$ m/s at what rate does the circle change when the radius is 10m?

- (a) $40m^2/s$ (b) $30m^2/s$
(c) $-\frac{30m^2}{s}$ (d) $-40m^2/s$

ANALYTICAL ABILITY AND LOGICAL REASONING

51. Decide which of the assumptions is implicit in the statement and choose your answer accordingly.

Statement:

Buy pure and natural honey of company 'X' - An advertisement in a newspaper.

Assumptions:

- I. Artificial honey can be prepared.
II. People do not mind paying more for pure and natural honey.

III. No other company supplies pure honey.

- (a) Only I is implicit (b) I and II are implicit
(c) I and III are implicit (d) All are implicit

52. Choose the conclusion which logically follows from the given statement(s)

Statement :

All scientist working in America are talented. Some Indian scientists are working in America.

Conclusions:

1. None of Indian scientist is talented.
2. Some talented Indian scientists have migrated to America.
3. All talented scientists are Indians.
4. Some indian scientists are talented

The conclusion(s) correctly drawn is/are

- (a) 1 only (b) 2 only
(c) 2 and 3 (d) 2 and 4

53. One New York publisher has estimated that 50000 to 60000 people in the united States want an anthology that includes the complete works of William Shakespeare. And what accounts for this renewed interest in Shakespeare? As scholars point out, his psychological insights into both male and female characters are amazing even today.

This paragraph best supports the statement that

- (a) Shakespeare 's characters are more interesting than fictional characters today
(b) People even today are interested in Shakespeare 's work because of the character
(c) Academic scholar are putting together an anthology of Shakespeare 's work
(d) New Yorkers have a renewed interested in the work of Shakespeare

54. A runs $1\frac{2}{3}$ times as fast as B. If A gives B start of 80m, how far must the winning post be so that A and B might reach it at the same time?

- (a) 200m (b) 300m
(c) 400m (d) 160m

55. Two men and three boys can do piece of work in ten days; while three men and two boys can do the same work in eight days. In how many days can two men and one boy do the work?

- (a) 12.5 (b) 9
(c) 9.5 (d) 8.5

56. How many numbers between 1 and 1000 are divisible by all numbers 2,3,4,5 and 6?

- (a) 16 (b) 32 (c) 17 (d) 33

57. There are 8436 steel balls, each with a radius of 1cm, stacked in a pile, with 1 ball on top, 3 balls in second layer, 6 in the third layer, 10 in the fourth and so on. the number of horizontal layer, in the pile is

- (a) 34 (b) 38 (c) 36 (d) 32



58. A body travels from A to B in 10s with a speed of 50 km/h and returns with a speed of 100km/h in 5s. The average speed and the average velocity for the whole journey is

- (a) $17.5 \text{ ms}^{-1}, 0 \text{ km/h}$ (b) $16.5 \text{ ms}^{-1}, 0 \text{ km/h}$
(c) $15.5 \text{ ms}^{-1}, 0 \text{ km/h}$ (d) $18.5 \text{ ms}^{-1}, 0 \text{ km/h}$

59. Find the odd one out.

- (a) DEHG (b) RSVU (c) JKNN (d) LMQP

60. Find out the wrong number in the following series.
15, 16, 34, 105, 424, 2124, 12756

- (a) 16 (b) 34 (c) 424 (d) 2124

61. My mother is twice as old as my brother. I am five years younger to my brother but three years older to my sister. If my sister is twelve years of age, how old is my mother?

- (a) 24 (b) 30 (c) 40 (d) 50

62. Find the missing term in the series.

3, 20, 63, 144, 275 _____.

- (a) 354 (b) 468 (c) 548 (d) 554

Directions (Q. Nos. 63 to 65) Questions are based on the following passage:

Nine individuals Z, Y, X, W, V, U, T, S and R are the only candidates who can serve on three committees A, B and C and each candidate should serve on exactly one of the committees. Committee A should consist of exactly one member more than Committee B. It is possible that there are no members of Committee C. Among Z, Y and X none can serve on Committee A. Among W, V and U none can serve on Committee B. Among T, S and R none can serve on Committee C.

63. In case T and Z are the individuals serving on Committee B, how many of the nine individuals should serve on Committee C?

- (a) 3 (b) 4 (c) 5 (d) 6

64. In case, T, S and X are the only individuals serving on committee B, the membership of Committee C should be

- (a) Z and Y (b) Z and W (c) Y and V (d) X and V

65. In case R is the only individual serving on Committee B, which among the following should serve on Committee A?

- (a) W and S (b) V and T
(c) U and S (d) T and S

66. In a city, 40% of the adults are illiterate while 85% of the children are literate. If the ratio of the adults to that of the children is 2: 3, then what per cent of the population is literate?

- (a) 20% (b) 25%
(c) 50% (d) 75%

67. There are 50 students admitted to a nursery class. Some students can speak only English and some can speak only Hindi. 10 students can speak both English and Hindi. If the number of students who can speak English is 21, then how many students can speak Hindi, how many can speak only Hindi, and how many can speak only English?

- (a) 21, 11 and 29 respectively
(b) 28, 18 and 22 respectively
(c) 37, 27 and 13 respectively
(d) 39, 29 and 11 respectively

68. The last digit of the number obtained by multiplying the numbers

$81 \times 82 \times 83 \times 84 \times 85 \times 86 \times 87 \times 88 \times 89$ will be

- (a) 0 (b) 9 (c) 7 (d) 2

69. It has been reported in recent years that a very large number of seats in the engineering colleges in the country remain vacant at the end of the admission session.

Which of the following may be the probable cause of the above effect?

- (a) There has been a considerable decrease in hiring of engineering graduates due to economic slowdown in the recent years
(b) Students have always preferred to complete graduation in three years time instead of four years for engineering
(c) The government has recently decided to provide post qualification professional training to all engineering graduates at its own cost
(d) None of the above

70. At what time, in minutes, between 3 O' clock and 4 O' clock, both the needles will coincide each other?

- (a) $5\frac{1}{2}$ " (b) $12\frac{4}{11}$ " (c) $13\frac{4}{11}$ " (d) $16\frac{4}{11}$ "

71. If $\log_3 2, \log_3(2^x - 5), \log_3(2^x - \frac{7}{2})$ are in arithmetic progression, then the value of x is equal to

- (a) 5 (b) 4 (c) 3 (d) 2

72. Let n be the number of different 5 digit numbers, divisible by 4 that can be formed with the digits 1, 2, 3, 4, 5 and 6, with no digit being repeated. What is the value of n?

- (a) 144 (b) 168 (c) 192 (d) 222

73. Let S be the set of integers x such that

- i. $100 \leq x \leq 200$
ii. x is odd and
iii. x is divisible by 3 but not by 7

How many elements does S contain?

- (a) 16 (b) 12 (c) 11 (d) 13





74. Two pipes A and B can fill a cistern in 37.5 min and 45 min, respectively. Both pipes are opened. The cistern will be filled in just half an hour, if the pipe B is turned off after

- (a) 5 min (b) 9 min (c) 10 min (d) 15 min

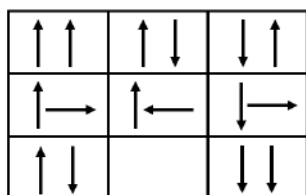
75. Three persons A, B and C wear shirts of Black, Blue and Orange colours (not necessarily in the order) and pants of Green, Yellow and Orange (not necessarily in that order). No person wore shirt and pant of the same colour. Further, it is given that,

1. A did not wear shirt of Black colour.
2. B did not wear shirt of Blue colour.
3. C did not wear shirt of Orange colour.
4. A did not wear the pant of Green colour.
5. B wore pant of Orange colour.

What were the colours of the pant and shirt worn by C respectively?

- (a) Orange and Black (b) Green and Blue
(c) Yellow and Blue (d) Yellow and Black

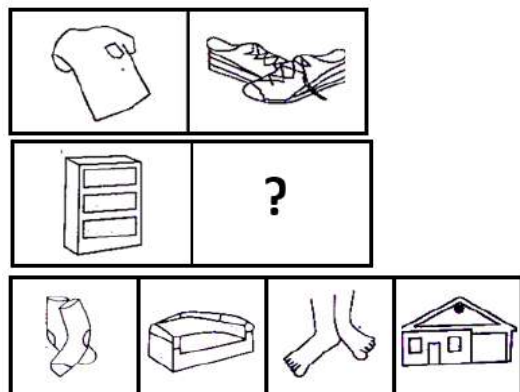
76.



Which one of the following figures fits into blank part of the above matrix?

- (a) (b)
(c) (d)

77. Choose the picture that would go in the empty box so that the two bottom pictures are related in the same way as the top two are related.



- (1) (2) (3) (4)
(a) 1 (b) 2 (c) 3 (d) 4

78. Decide which of the given conclusions logically follow from the given statement(s).

Statements Some codes are secrets.

All secrets are puzzles.

Conclusions:

- I. All secrets being codes is a possibility.
 - II. Atleast some puzzles are codes.
- (a) Only conclusion I is true
(b) Only conclusion II is true
(c) Either conclusion I or II is true
(d) Both conclusions I and II are true

79. Decide which of the given conclusions logically follow from the given statement(s).

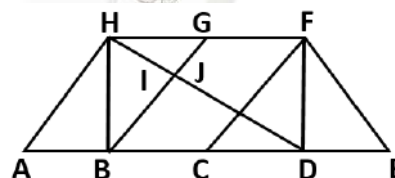
Statements:

Some metals are alloys. No metal is a stone.

Conclusions

- I. No alloy is stone.
 - II. Atleast some alloys are metals.
- (a) Neither conclusion I nor II true
(b) Only conclusion II is true
(c) Only conclusion I is true
(d) Both conclusions I and II are true

80. Find the number of triangles in the given picture.



- (a) 10 (b) 12 (c) 14 (d) 16

81. Decide which of the given conclusions logically follows from the given statement(s)

Statements:

All mangoes are golden in colour. No golden coloured things are cheap.

Conclusions:

- (I) All mangoes are cheap.
 - (II) Golden-coloured mangoes are not cheap.
- (a) only conclusions I follows
(b) only conclusions II follows
(c) Either I or II follows
(d) Neither I nor II follows

82. By selling an article, what is the profit percent gained?

- I. 5% discount is given on list price.
- II. If discount is not given, 20% profit is gained.
- III The cost price of the articles is Rs.5000.

- (a) I and II (b) II and III
(c) I and III (d) All of I, II and III



83. Decide which of the given conclusions logically follows from the given statement(s).

Statements:

All politicians are honest. All honest are fair.

Conclusions :

- I. Some honest are politicians
- II. No honest is politicians
- III. Some fair are politicians
- IV. All fair are politicians

- (a) None follows
- (b) I and IV follows
- (c) I and II follows
- (d) I and III follows

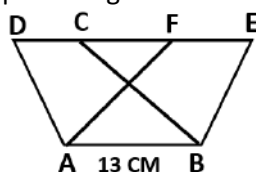
84. Forty students watched films A, B and C over a week. Each student watched either only one film or all three. Thirteen students watched film A, sixteen students watched film B and nineteen students watched film C. How many students watched all three films?

- (a) 0
- (b) 2
- (c) 4
- (d) 8

85. Two bus tickets from city A to B and three tickets from city A to C cost Rs.77, but three tickets from city A to B and two tickets from city A to C cost Rs.73. What are the fares for cities B and C from A?

- (a) Rs. 4, Rs. 23
- (b) Rs. 13, Rs. 17
- (c) Rs. 15, Rs. 14
- (d) Rs. 17, Rs. 13

86. In the figure given below, if the area of parallelogram ABCD is 208cm^2 , what is the height of the parallelogram ABEF?



- (a) 15 cm
- (b) 15.5 cm
- (c) 16 cm
- (d) 16.5 cm

87. If the first and the third letters in the group DISTRIBUTION are interchanged and also the second and the fourth letter, the fifth and the seventh and so on, then which of the following would be the seventh letter from the left?

- (a) U
- (b) R
- (c) B
- (d) T

88. In the below series, you will be looking at the letter pattern, diagram pattern or number pattern. Fill in the blank in the end of the series. JAK, KBL, LCM, MDN, _____.

- (a) OEP
- (b) NEO
- (c) MEN
- (d) PFQ

89. A caterpillar crawls up a pole of 75 inches high, starting from the ground. Each day, it crawls up 5 inches and each night it slides down 4 inches. When will it reach the top of the pole?

- (a) End of 70 days
- (b) End of 71 days
- (c) End of 72 days
- (d) End of 73 days

90. The time on the watch is quarter to three. If the minute hand points of North-East, in which direction does the hour hand point?

- (a) North-West
- (b) South-West
- (c) South-East
- (d) North-East

91. What is the minimum form of Karnaugh map shown below? (Assume that x denotes a don't care term)

ab \ cd	00	01	11	10
00	1	X	X	1
01	X			1
11				
10	1			X

- (a) $\bar{b}\bar{d}$
- (b) $\bar{b}\bar{d} + \bar{b}\bar{c}$
- (c) $\bar{b}\bar{d} + \bar{a}\bar{d}\bar{c}\bar{d}$
- (d) $\bar{b}\bar{d} + \bar{b}\bar{c} + \bar{c}\bar{d}$

92. In order to store floating numbers in computers using the normalized representation and 32-bit single precision, the number of bits used for exponent and fraction are _____, _____ respectively.

- (a) 11, 21
- (b) 16, 15
- (c) 16, 16
- (d) 8, 23

93. Let the memory access time is 10 milliseconds and cache access time is 10 microseconds. Assume the cache hit ratio 15%. The effective memory access time is

- (a) 18.51 milliseconds
- (b) 18 milliseconds
- (c) 19 microseconds
- (d) 20 milliseconds

94. What are the values of R_1 and R_2 respectively in the expression

$$(235)_{R_1} = (565)_{10} = (1065)_{R_2}?$$

- (a) 8, 16
- (b) 16, 8
- (c) 8, 12
- (d) 16, 12

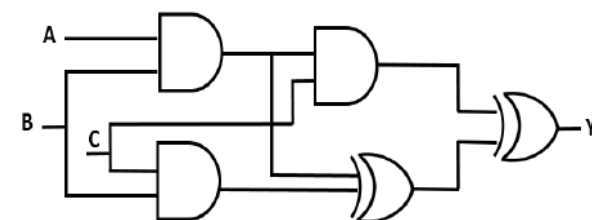
95. If a signal passing through a gate is inhibited by sending a low into one of the inputs and the output is high, the gate is

- (a) NOR
- (b) NAND
- (c) AND
- (d) OR

96. Given, $\sqrt{(224)_r} = (13)_r$, the value of the radix is

- (a) 10
- (b) 8
- (c) 5
- (d) 6

97. The output of the combinational circuit given below is



- (a) $A + B + C$
- (b) $A(B + C)$
- (c) $B(C + A)$
- (d) $C(A + B)$

98. Which one of the following expressions does NOT represent exclusive NOR of x and y?

- (a) $xy + \bar{x}\bar{y}$
- (b) $x \oplus y$
- (c) $\bar{x} \oplus y$
- (d) $\bar{x} \oplus \bar{y}$

99. What type of error are not detected by assemblers?

- (a) Syntax error
- (b) Run-time error
- (c) Logical error
- (d) All of these



100. Consider the equation $(43)_x = (y3)_8$, where x and y are unknown. The number of possible solutions is
- (a) 4 (b) 5
(c) 6 (d) 7

GENERAL ENGLISH

101. Find the synonym that is most nearly similar in meaning to the word: DEBACLE.
- (a) Catastrophe (b) Dandy
(c) Corker (d) Opulence
102. Choose the phrasal verb the means "To spend time doing unimportant things-instead of doing necessary things".
- (a) fake out (b) faff about
(c) fade out (d) fall apart
103. Which of the following is close to the meaning of the word, IMPOLITIC?
- (a) Impolite (b) Intolerant
(c) Incongruous (d) Injudicious
104. Choose an option to replace the phrase given in bold. The bank has hired a consultant who **will look into** any issues which arise during the merger.
- (a) is looking over (b) will be looked over
(c) will look out (d) no correction required
105. The sentences given below when properly sequenced, form a coherent paragraph. Each sentence is labeled with a letter. Choose the most logical order of sentences from the given choices to construct a coherent paragraph.
- P: It refers to those times when there is no war or fight among nations.
Q: Many people argue that war is a necessary evil and the only process of solving international disputes.
R: Peace means complete freedom from disturbance.
S: Once the world starts believing that wars are unnecessary, they decide to abolish war completely.
T: But in reality, war happens only because people are not ready to settle disputes in a peaceful manner.
- (a) TSRQP (b) RTQPS
(c) SPTQR (d) QTRPS

Directions (Q. Nos. 106 to 108) The following passage and answer the questions based on it. I do not wish to cast aspersions on the corporate takeover. On the contrary, it is a key facilitator of creative destruction and doubtless the most effective remaining means by which shareholder voices can mold a corporation. But while change in management is often necessary, you cannot effectively run a corporation with differing authoritative voices espousing opposing corporate goals. It has to be one or the other. If the board is riven with conflicting interests, corporate governance will suffer. If directors cannot agree with the CEO's strategy, they should replace him. Corporate

dissonance, of course, is unavoidable in periods of transition. But is not a value to be pursued for its own sake. A cacophony produces only red ink.

106. What according to the author facilitates creative destruction?
- (a) Corporatism (b) Modernism
(c) Terrorism (d) Cosmopolitanism
107. Which of the following words used in the passage reinforces the idea contained in the word, 'DISSONANCE'?
- (a) Aspersions (b) Cacophony
(c) Strategy (d) Transition
108. What according to the author makes corporate governance suffer?
- (a) Conflicting strategies of executives
(b) Conflicting interests of customers
(c) Conflicting interests of statesmen
(d) Conflicting interests of the Board of Directors
109. Fill in the blank choosing the correct word. The vote will have to be laid until next week.
- (a) on (b) in (c) over (d) from
110. Fill in the blank choosing the correct question tag. He has done his duty,.....
- (a) Shouldn't he? (b) Hasn't he?
(c) Won't he? (d) Has he?
111. Choose one of the options that is most nearly same as meaning of the word Epitome.
- (a) Final verdict (b) Climax
(c) Essence (d) Tombstone
112. Which collocation goes with the word AWARE?
- (a) rightly (b) fully (c) nearly (d) exactly
113. Change the speech : "If you don't keep quite I shall shoot you", he said to her in a calm voice.
- (a) He warned her to shoot if she didn't keep quite calmly
(b) He said calmly that I shall shoot you if you don't be quite
(c) He warned her calmly that he would shoot her if she didn't keep quite
(d) Calmly he warned her that be quite or else he will have to shoot her
114. Choose the correct spelling for the word given below.
- (a) Ceiling (b) Cealing
(c) Ceiling (d) Ceeling
115. Select the pair of words, which are related in the same way as the capitalised words are related to each other. BUTTERFLY: FREEDOM
- (a) Frog: Tadpole (b) Self reliant: Buoyant
(c) Alga: Lichens (d) Chicken: Rooster



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116. Which of the following is the closest in meaning to the word, CLOWN?

- (a) Idiot (b) Dunce
- (c) Don (d) Jester

117. Identify the underlined part which has an error. He couldn't cope up with the hot climate of the place any longer.

- (a) couldn't (b) cope up
- (c) climate (d) any longer

118. The _____ pittance the widow receives from the government cannot keep her from poverty.

- (a) Meagre (b) Indulgent
- (c) Meticulous (d) Magnanimous

119. The other boys or Henry..... to blame.

- (a) is (b) are (c) were (d) will

120. In the question, the word 'Echo' is usual in four different ways, numbered A to D.

Choose the option in which the usage of the word is inappropriate.

- (a) The crowd cheered to the echo in the foot ball match
- (b) I started to feel nostalgic as I stood in the old neighbourhood that echoed with my childhood
- (c) A loud thunder echo hit the tallest building
- (d) My thoughts echoed with the sounds of spring

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MAARULA MATHEM –E LEARNING

ANSWER KEY

- | | | | | | |
|---------|---------|---------|---------|----------|----------|
| 1. (a) | 21. (a) | 41. (a) | 61. (c) | 81. (b) | 101. (a) |
| 2. (b) | 22. (b) | 42. (a) | 62. (b) | 82. (d) | 102. (b) |
| 3. (a) | 23. (d) | 43. (b) | 63. (b) | 83. (d) | 103. (d) |
| 4. (d) | 24. (d) | 44. (b) | 64. (a) | 84. (d) | 104. (d) |
| 5. (a) | 25. (c) | 45. (c) | 65. (d) | 85. (b) | 105. (d) |
| 6. (c) | 26. (b) | 46. (b) | 66. (d) | 86. (c) | 106. (a) |
| 7. (d) | 27. (a) | 47. (c) | 67. (d) | 87. (b) | 107. (b) |
| 8. (b) | 28. (d) | 48. (c) | 68. (a) | 88. (b) | 108. (d) |
| 9. (c) | 29. (c) | 49. (b) | 69. (a) | 89. (b) | 109. (c) |
| 10. (d) | 30. (d) | 50. (d) | 70. (d) | 90. (b) | 110. (b) |
| 11. (b) | 31. (a) | 51. (a) | 71. (c) | 91. (d) | 111. (c) |
| 12. (c) | 32. (d) | 52. (d) | 72. (c) | 92. (d) | 112. (b) |
| 13. (a) | 33. (c) | 53. (b) | 73. (d) | 93. (a) | 113. (c) |
| 14. (b) | 34. (b) | 54. (a) | 74. (b) | 94. (b) | 114. (c) |
| 15. (b) | 35. (b) | 55. (a) | 75. (b) | 95. (b) | 115. (b) |
| 16. (c) | 36. (b) | 56. (a) | 76. (b) | 96. (c) | 116. (d) |
| 17. (b) | 37. (b) | 57. (c) | 77. (b) | 97. (c) | 117. (b) |
| 18. (a) | 38. (a) | 58. (d) | 78. (d) | 98. (d) | 118. (a) |
| 19. (d) | 39. (a) | 59. (d) | 79. (b) | 99. (c) | 119. (a) |
| 20. (d) | 40. (b) | 60. (d) | 80. (c) | 100. (b) | 120. (c) |



SOLUTION

MATHS

1. (a) $S: x^2 + y^2 + 10x - 12y + 51 = 0$,
 $L: 3y + x = 3$
 $x = 3 - 3y$
 $\Rightarrow 9 + 9y^2 - 18y + y^2 + 30 - 30y - 12y + 51 = 0$
 $\Rightarrow 10y^2 - 60y + 90 = 0$
 $\Rightarrow y^2 - 6y + 9 = 0$
 $\Rightarrow (y - 3)^2 = 0$
 $\Rightarrow y = 3$
 $\Rightarrow x = 3 - 3(3)$
 $\Rightarrow x = -6$
 $\therefore P(-6, 3)$

2. (b)
 $2 \sin 3x \cos 2x - \sin 3x = 0$
 $\Rightarrow \sin 3x(2 \cos 2x - 1) = 0$
 $\Rightarrow \sin 3x = 0; \cos 2x = \frac{1}{2}$
 $\sin 3x = 0 \Rightarrow 3x = 0, \pi, 2\pi, 3\pi$
Hence, $x = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi$
And $\cos 2x = \frac{1}{2} \Rightarrow 2x = \frac{\pi}{3}, \frac{5\pi}{3}$
Hence, $x = \frac{\pi}{6}, \frac{5\pi}{6} \Rightarrow x = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi$ and $\frac{\pi}{6}, \frac{5\pi}{6}$
 $\Rightarrow$ Number of solutions = 6.

3. (a)
 $f(x) = \sec A + \sec B + \sec C$
 $\therefore$ For any triangle $\Rightarrow \cos A + \cos B + \cos C = \frac{3}{2}$
 $A + B + C = \pi \therefore A, B, C \in 1^{\text{st}} \text{ quad}$
so all angles will be positive $\Rightarrow AM \geq GM$
 $\Rightarrow \frac{\cos A + \cos B + \cos C}{3} \geq (\cos A \cos B \cos C)^{1/3}$
 $\Rightarrow \left(\frac{3/2}{3}\right)^3 \geq \cos A \cos B \cos C$
 $\Rightarrow \cos A \cos B \cos C \leq 1/8 \Rightarrow \sec A \sec B \sec C \leq 8$
 $\frac{\sec A + \sec B + \sec C}{3} \geq (\sec A \sec B \sec C)^{1/3}$
 $\Rightarrow \sec A + \sec B + \sec C \geq 3 \times 2$
 $\Rightarrow \sec A + \sec B + \sec C \geq 6$

4. (d)
 $P: \sin \theta - \cos \theta = \sqrt{2} \cos \theta$
 $\Rightarrow \sin \theta = (\sqrt{2} + 1) \cos \theta \Rightarrow \tan \theta = \sqrt{2} + 1$
 $Q: \sin \theta + \cos \theta = \sqrt{2} \sin \theta \Rightarrow \cos \theta = (\sqrt{2} - 1) \sin \theta$
 $\Rightarrow \tan \theta = \frac{1}{\sqrt{2} - 1} = \sqrt{2} + 1 \Rightarrow P = Q$

5. (a)
 $\frac{\tan x}{2} = \frac{\tan y}{3} = \frac{\tan z}{5} = k$
 $\Rightarrow \tan x = 2k, \tan y = 3k$ and $\tan z = 5k$
 $x + y = \pi - z$
 $\Rightarrow \tan(x + y) = -\tan z \Rightarrow \frac{\tan x + \tan y}{1 - \tan x \tan y} = -\tan z$
 $\Rightarrow \tan x + \tan y + \tan z = \tan x \tan y \tan z$
 $\Rightarrow 2k + 3k + 5k = (2k)(3k)(5k)$
 $\Rightarrow 10k = 30k^3 \Rightarrow k^2 = \frac{1}{3}$
and $(\tan x)^2 + (\tan y)^2 + (\tan z)^2 = 38k^2 = \frac{38}{3}$

6. (c)

$$S_1: x^2 + y^2 - 2ax + c^2 = 0, x^2 + y^2 - 2by + c^2 = 0$$

$$C_1(a, 0) \quad C_2(0, b)$$

$$r_1 = \sqrt{a^2 - c^2} \quad r_2 = \sqrt{b^2 - c^2}$$

$$r_1 = \sqrt{a^2 - c^2} \quad r_2 = \sqrt{b^2 - c^2}$$

$$S_1 \text{ and } S_2 \text{ touches externally}$$

$$\therefore C_1 C_2 = r_1 + r_2$$

$$\sqrt{a^2 + b^2} = \sqrt{a^2 - c^2} + \sqrt{b^2 - c^2}$$

$$\Rightarrow a^2 + b^2 = a^2 - c^2 + b^2 - c^2 + 2\sqrt{a^2 - c^2} \sqrt{b^2 - c^2}$$

$$\Rightarrow 2c^2 = 2\sqrt{a^2 - c^2} \sqrt{b^2 - c^2}$$

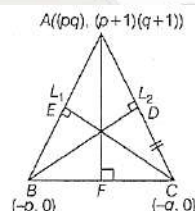
$$\Rightarrow c^4 = (a^2 - c^2)(b^2 - c^2)$$

$$\Rightarrow c^4 = a^2 b^2 - a^2 c^2 - b^2 c^2 + c^4$$

$$\Rightarrow a^2 b^2 = a^2 c^2 + b^2 c^2$$

$$\Rightarrow \frac{1}{c^2} = \frac{1}{a^2} + \frac{1}{b^2}$$

7. (d) $L_1: (1 + p)x - py + p(1 + p) = 0$
 $L_2: (1 + q)(x) - qy + q(1 + q) = 0 \Rightarrow L_3: y = 0$
Point of intersection : $y = 0$ in L_1
 $x = -p; B(-p, 0) \Rightarrow y = 0$ in L_2
 $x = -q; C(-q, 0)$
Point of intersection : $y = 0$ in L_1
Point of intersection L_1 and $L_2: x = pq, y = (1 + p)(1 + q)$



Altitude from A to BC is $x = pq$
Altitude from B to AC is $y = \frac{-q}{1+q}(x + q)$
Solving the two lines, we get $x = pq$ and $y = -pq$
 $\therefore$ Locus of orthocenter is $x + y = 0$.
This is a straight line.

8. (b) Hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$

Equation of its tangent $bx \sec \theta - ay \tan \theta = ab$
Here $b = 2, a = 3$
Hence equation of tangent
 $2x \sec \theta - 3y \tan \theta = 6$

Equation of circle $x^2 + y^2 - 8x = 0$, center $(4, 0)$, radius = 4
For common tangent perpendicular from center of circle to given line it should be equal to radius.

$$\left| \frac{8 \sec \theta - 6}{\sqrt{4 \sec^2 \theta + 9 \tan^2 \theta}} \right| = 4$$

$$\Rightarrow (4 \sec \theta - 3)^2 = 4(4 \sec^2 \theta + 9 \tan^2 \theta)$$

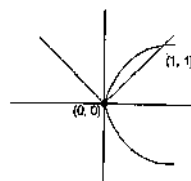
$$\Rightarrow 12 \sec^2 \theta + 8 \sec \theta - 15 = 0$$

$$\Rightarrow \sec \theta = -\frac{3}{2}, \frac{5}{6} \text{ But } \sec \theta \neq \frac{5}{6}$$

If $\sec \theta = -\frac{3}{2}$ then $\tan \theta = \frac{\sqrt{5}}{2}$

Put these values in equation of tangent, we get
 $2x - \sqrt{5}y + 4 = 0$

9. (c) Intersection point $\Rightarrow |x|^2 = x$
 $x^2 = x$



Point of intersections are $(0, 0)$ and $(1, 1)$



$$A = \int_0^1 (\sqrt{x} - |x|) dx$$

$$A = \int_0^1 (\sqrt{x} - x) dx$$

$$= \left[\frac{2x^{3/2}}{3} - \frac{x^2}{2} \right]_0^1$$

$$= \frac{2}{3} - \frac{1}{2} = \frac{1}{6} \text{ sq. units}$$

$$A = \frac{1}{6} \text{ sq. units}$$

10. (d) Equation of the line perpendicular to $x - 2y = 1$

Let equation of line be, $2x + y = \lambda$

It passes through $(1,1)$: $\lambda = 3$

$$\therefore L: 2x + y = 3$$

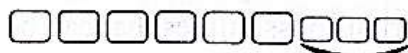
$$\Rightarrow y = -2x + 3$$

11. (b) $A^2 = \begin{bmatrix} 0 & 5 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 5 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

So, A^2 and all higher powers of matrix A will be null matrix.

$$f(A) = 1 + A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 5 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 5 \\ 0 & 1 \end{bmatrix}$$

12. (c)



small boxes

$$\text{Required number of arrangements} = {}^6P_5 \times 4! \\ = 720 \times 24 = 17280$$

13. (a) $y = \frac{1-x}{1+x} \Rightarrow y + yx = 1 - x$

$$\Rightarrow yx + x = 1 - y$$

$$\Rightarrow x(y + 1) = 1 - y$$

$$\Rightarrow x = \frac{1-y}{1+y}$$

Now, interchange x and y

$$f^{-1}(x) = y = \frac{1-x}{1+x}$$

14. (b) Given that, Student council has total numbers = 10

To select one President, One Vice-President, One Secretary, One Joint-Secretary and two Executive Committee members from 10 members.

Firstly, we have to select 6 members from 10 members. Which can be

done in total number of ways = ${}^{10}C_6$

But there are two Executive repeated members.

$\therefore$ That number of ways will be = $\frac{6!}{2!}$

Hence, total number of ways in which we can appoint

three members = ${}^{10}C_6 \times \frac{6!}{2!}$

$$= \frac{10!}{4!6!} \times \frac{6!}{2!} \left[\because {}^nC_r = \frac{n!}{(n-r)!r!} \right]$$

$$= \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5}{2}$$

$$= 10 \times 9 \times 8 \times 7 \times 3 \times 5 = 75600$$

15. (b) Only exactly one subject

$$[n(M) + n(L) + n(B)] - 2[n(M \cap L) + n(L \cap B) + n(B \cap M)] + 3n(M \cap L \cap B)$$

$$= [32 + 38 + 30] - 2[7 + 10 + 8] + 3[5]$$

$$= 100 - 2[25] + 15 = 100 - 50 + 15 = 65$$

16. (c) According to question, following cases arises nine digit number (less than 2×10^8) having only digits 1 and 2 = 2^8 [As

first place as only one choice i.e., number 1 and rest have two choices]

Eight digit number (less than 2×10^8) having only one digit 1 and 2 = 2^8

Similarly, seven digit numbers = 2^7 .

Six digit numbers = 2^6 .

Five digit numbers = 2^5 , Four digit numbers = 2^4 , three digit numbers = 2^3 ,

Two digit numbers = 2^2 , One digit numbers = 2

So, numbers of all such natural numbers.

$$2 + 2^2 + 2^3 + 2^4 + 2^5 + 2^6 + 2^7 + 2^8 + 2^8$$

$$= 2 + 4 + 8 + 16 + 32 + 64 + 128 + 256 + 256 = 766$$

17. (b) $\lim_{x \rightarrow \infty} \left(1 + \frac{a}{x} + \frac{b}{x^2}\right)^{2x} = e^2$ [1 $^\infty$ form]

$$= e^{\lim_{x \rightarrow \infty} \left[\frac{a+b}{x} + \frac{b}{x^2}\right] 2x} = e^2$$

$$= e^{\lim_{x \rightarrow \infty} [a + \frac{b}{x}] 2} = e^2$$

$$e^{2a} = e^2 \Rightarrow a = 1, b \in R$$

18. (a) $f(x) = \sin^5 x + \sin^3 x \Rightarrow f(-x) = -\sin^5 x - \sin^3 x$

$$f(x) + f(-x) = \sin^5 x + \sin^3 x - \sin^5 x - \sin^3 x = 0$$

$$g(x) = \cos^5 x + \sin^3 x \Rightarrow g(-x) = \cos^5 x - \sin^3 x$$

$$g(x) + g(-x) = 2\cos^5 x$$

$$\text{Then, find } \int_0^{\pi/2} [f(x) + f(-x)][g(x) + g(-x)] dx$$

$$= \int_0^{\pi/2} 0 \times 2\cos^5 x dx = 0$$

19. (d) $\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}} = \left(\frac{dy}{dx}\right)^{-1}$

$$\frac{d}{dy} \left(\frac{dx}{dy}\right) = \frac{d}{dy} \left(\frac{dy}{dx}\right)^{-1} \times \left(\frac{dx}{dy}\right)$$

$$\frac{d^2x}{dy^2} = -\left(\frac{dy}{dx}\right)^{-2} \times \frac{d^2y}{dx^2} \times \left(\frac{dy}{dx}\right)^{-1}$$

$$\left[\frac{d^2x}{dy^2} = -\left(\frac{dy}{dx}\right)^{-3} \times \left(\frac{d^2y}{dx^2}\right)\right]$$

20. (d) let $u = \log_{10} x$, $v = \log_x 10$

$$u = \frac{\log x}{\log 10}, v = \frac{\log 10}{\log x}$$

$$\frac{du}{dx} = \frac{1}{\log 10} \frac{d}{dx} (\log x), \frac{dv}{dx} = \log_{10} \frac{d}{dx} \left(\frac{1}{\log x}\right)$$

$$\frac{du}{dx} = \frac{1}{\log 10} \left(\frac{1}{x}\right), \frac{dv}{dx} = \log_{10} \frac{d}{dx} (\log x)^{-1}$$

$$\frac{du}{dx} = \frac{1}{x \log 10}, \frac{dv}{dx} = -\log_{10} \left[(\log x)^{-2} \frac{1}{x}\right]$$

$$\frac{du}{dx} = \frac{1}{x \log 10}, \frac{dv}{dx} = \frac{-\log 10}{x \cdot (\log x)^2}$$

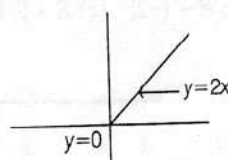
$$\frac{du}{dv} = \frac{du/dx}{dv/dx}$$

$$= \frac{1/x \log 10}{\frac{-\log 10}{x \cdot (\log x)^2}}$$

$$= \frac{1}{x \log 10} \times \frac{x (\log x)^2}{-\log 10}$$

$$\left[\frac{du}{dv} = -\frac{(\log x)^2}{-(\log 10)^2} (d)\right]$$

21. (a)





$$f(x) = x + |x| = \begin{cases} 2x & x \geq 0 \\ 0 & x < 0 \end{cases}$$

$f(x)$ is Continuous everywhere $(-\infty, \infty)$

22. (b)

$$\begin{aligned} &|a-b|^2 + |b-c|^2 + |c-a|^2 \\ &= 2(|a|^2 + |b|^2 + |c|^2) - 2(a \cdot b + b \cdot c + c \cdot a) \\ &= 6 - 2(a \cdot b + b \cdot c + c \cdot a) \\ &= 6 - ((a+b+c)^2 - (|a|^2 + |b|^2 + |c|^2)) \\ &= 9 - (a + b + c)^2 \\ &9 - \{a + b + c\} \leq 9 \end{aligned}$$

23. (d) $a = b + \lambda c$

$$\begin{aligned} \alpha \hat{i} + 2\hat{j} + \beta \hat{k} &= \hat{i} + \hat{j} + \lambda \hat{j} + \lambda \hat{k} \\ \alpha \hat{i} + 2\hat{j} + \beta \hat{k} &= \hat{i} + (1 + \lambda)\hat{j} + \lambda \hat{k} \\ 1 + \lambda &= 2 \\ \lambda &= 1, \alpha = 1 \text{ so, } \beta = 1 \\ \alpha &= \beta = 1 \end{aligned}$$

24. (a) $F_{net} = 4\hat{i} - 3\hat{j} + 7\hat{k} - 2\hat{i} + 2\hat{j} - 8\hat{k} = 2\hat{i} - \hat{j} - \hat{k}$

$$r = 2\hat{i} + 5\hat{j} - 6\hat{k} - 5\hat{i} - 7\hat{j} - \hat{k} = -3\hat{i} - 2\hat{j} - 7\hat{k}$$

$$W = F \cdot r$$

$$W = 3 \text{ units.}$$

25. (c) Here, velocity vector of bird is

$$v = 10\hat{i} + 6\hat{j} + \hat{k}$$

$$\text{Or } |v_x| = 10 \text{ km/h}$$

$$|v_y| = 6 \text{ km/h}$$

$$|v_z| = 1 \text{ km/h}$$

Initial position vector of the bird

$$OA = \hat{i} + 2\hat{j} + 3\hat{k}$$

When bird reached a point in space that is 13m high from ground then final position vector of the bird will be

$$OB = \hat{i} + 2\hat{j} + 13\hat{k}$$

$$AB = OB - OA = 10\hat{k}$$

$$\therefore |AB| = 10 \text{ m}$$

Time taken by bird to cover a distance of 10m along the vertical is-

$$t = \frac{AB}{V_z} = \frac{10}{1 \times \frac{5}{18}} = \frac{10 \times 18}{5} = 36 \text{ sec}$$

26. (b)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\frac{5}{6} = P(A) + \frac{1}{2} - \frac{1}{3} \Rightarrow P(A) = \frac{2}{3}$$

If A and B are independent to each other

$$P(A \cap B) = P(A) \times P(B) = \frac{2}{3} \times \frac{1}{2} = \frac{1}{3}$$

Hence, A and B are independent events.

27. (a) $a_1 a_2 a_3 \dots a_n = C$

$$AM \geq GM$$

$$\frac{a_1 + a_2 + \dots + a_n + 2an}{n} \geq \sqrt[n]{a_1 a_2 a_3 \dots a_n 2(an)}$$

$$\frac{a_1 + a_2 + a_3 + \dots + a_{n-1} + 2a_n}{n} \geq (2c)^{1/n}$$

$$[a_1 + a_2 + a_3 \dots a_{n-1} + 2a_n = n(2c)^{1/2}] \text{ will be minimize}$$

28. (d) Using Vieta's theorem for

$$x^3 - 3x^2 + 3x + 7 = 0 \text{ has roots } a, b, c$$

$$a + b + c = 3, ab + bc + ca = 3, abc = -7$$

$$\begin{vmatrix} 2bc - a^2 & c^2 & b^2 \\ c^2 & 2ac - b^2 & a^2 \\ b^2 & a^2 & 2ab - c^2 \end{vmatrix}$$

$$= \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \begin{vmatrix} -a & c & b \\ -b & a & c \\ -c & b & a \end{vmatrix} = - \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \begin{vmatrix} a & c & b \\ b & a & c \\ c & b & a \end{vmatrix}$$

$$= \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}^2$$

$$= (3abc - a^3 - b^3 - c^3)^2$$

We know that,

$$(a + b + c)^3 = a^3 + b^3 + c^3 + 3(a + b + c)(ab + bc + ca) - 3abc$$

$$\Rightarrow 27 = a^3 + b^3 + c^3 + 3(3)(3) + 3 \times 7$$

$$\Rightarrow a^3 + b^3 + c^3 = -21$$

$$\text{So, } (3abc - a^3 - b^3 - c^3)^2 = [3(-7) + 21]^2 = 0$$

29. (c) We should know,

$$(1 - 2x + 3x^2 - 4x^3 + \dots \infty) = (1 + x)^{-2}$$

$$\Rightarrow (1 - 2x + 3x^2 - 4x^3 + \dots \infty)^{-n}$$

$$[(1 + x)^{-2}]^{-n} = (1 + x)^{2n}$$

General term of $(1 + x)^{2n}$ is

$$T_{r+1} = {}^{2n}C_r x^r$$

For coefficient of x^n

$$r = n = {}^{2n}C_n = \frac{(2n)!}{(2n-n)!n!} = \frac{(2n)!}{(n!)(n!)}$$

$$= \frac{(2n)!}{(n!)^2}$$

30. (d) $\alpha + \beta = p; \alpha\beta = r; \frac{\alpha}{2} + 2\beta = q; \frac{\alpha}{2} \cdot 2\beta = r$

$$\Rightarrow \alpha\beta = r$$

$$\alpha + \beta = p \quad \dots\dots(i)$$

$$\alpha + 4\beta = 2q \quad \dots\dots(ii)$$

On subtracting Eq. (ii) from Eq. (i), we get

$$3\beta = 2q - p \Rightarrow \beta = \frac{(2q-p)}{3}$$

$$\alpha + \beta = p$$

$$\Rightarrow \alpha = p - \beta = p - \frac{(2q-p)}{3} = \frac{3p-2q+p}{3}$$

$$\Rightarrow \alpha = \frac{4p-2q}{3} = \frac{2}{3}(2p-q)$$

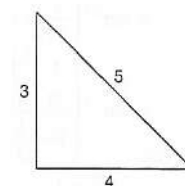
$$\Rightarrow r = \alpha + \beta = \frac{2}{3}(2p-q) \cdot \frac{1}{3}(2q-p) = \frac{2}{9}(2p-q)(2q-p)$$

31. (a)

$$\cot\left(\operatorname{cosec}^{-1}\frac{5}{3} + \tan^{-1}\frac{2}{3}\right) = \cot\left(\tan^{-1}\frac{3}{4} + \tan^{-1}\frac{2}{3}\right)$$

$$= \cot\left[\tan^{-1}\left(\frac{\frac{3}{4} + \frac{2}{3}}{1 - \frac{6}{12}}\right)\right]$$

$$= \cot\left(\tan^{-1}\frac{9+8}{6}\right)$$



$$= \cot\left(\tan^{-1}\frac{17}{6}\right) = \cot\left(\cot^{-1}\frac{6}{17}\right) = \frac{6}{17}$$

32. (d)

$$3\sin(\overline{\theta + \alpha} + \alpha) = \sin(\theta + \alpha - \alpha)$$



$$\begin{aligned} &\Rightarrow 3[\sin(\theta + \alpha)\cos\alpha + \cos(\theta + \alpha)\sin\alpha] \\ &= \sin(\theta + \alpha)\cos\alpha - \cos(\theta + \alpha)\sin\alpha \\ &\Rightarrow \sin(\theta + \alpha)\cos\alpha = -2\cos(\theta + \alpha)\sin\alpha \\ &\Rightarrow 2\tan\alpha + \tan(\theta + \alpha) = 0 \end{aligned}$$

Alternate solution

$$\sin\theta = 3\sin(\theta + 2\alpha)$$

$$\text{Let } \theta = \alpha = 0$$

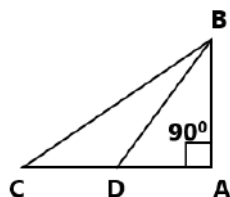
$$\Rightarrow 0 = 0; \text{ verified. } \Rightarrow \tan(\alpha + \theta) + 2\tan\alpha = 0$$

33. (c) $BC^2 = BA^2 + (2AD)^2$

$$BD^2 = BA^2 + AD^2$$

$$BC^2 - BD^2 = BA^2 + 4AD^2 - BA^2 - AD^2$$

$$BC^2 - BD^2 = 3AD^2$$



34. (b) Area of OPB = 2 (area of OAP)

$$\therefore xy - \int_0^x y dx = 2 \int_0^x y dx$$

$$\Rightarrow xy = 3 \int_0^x y dx$$

Differentiation with respect to x , we get

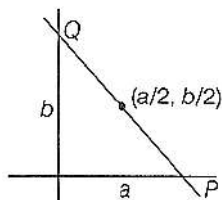
$$y + x \frac{dy}{dx} = 3y \Rightarrow \frac{dy}{y} = 2 \frac{dx}{x} \Rightarrow \log y$$

$$= 2 \log x + \log c \Rightarrow \log y = \log cx^2 \Rightarrow y = cx^2$$

This is an integrating yields $y = cx^2$.

Hence, it is a parabola.

35. (b) Let equation be $\frac{x}{a} + \frac{y}{b} = 1$



$$(4, 2) \text{ lies on } PQ \Rightarrow \frac{4}{a} + \frac{2}{b} = 1$$

In the given diagram circumcentre will be $(\frac{a}{2}, \frac{b}{2})$

We have to find locus of circumcentre which we will assume as (h, k)

$$\text{Hence, } a = 2h, b = 2k$$

$$\frac{4}{2h} + \frac{2}{2k} = 1 \Rightarrow \frac{2}{h} + \frac{1}{k} = 1 \text{ or } \frac{2}{x} + \frac{1}{y} = 1$$

36. (b) Let $x = \frac{1}{\sqrt{2}} \sqrt{1 - \frac{1}{\sqrt{2}} \sqrt{1 - \frac{1}{\sqrt{2}} \sqrt{1 - \frac{1}{\sqrt{2}}}}}$

$$\Rightarrow x = \frac{1}{\sqrt{2}} \sqrt{1 - x} \Rightarrow x^2 = \frac{1}{2}(1 - x), \text{ We know } x > 0$$

$$\Rightarrow 2x^2 = 1 - x \Rightarrow 2x^2 + x - 1 = 0$$

$$\Rightarrow 2x^2 + 2x - x - 1 = 0 \Rightarrow (2x - 1)(x + 1) = 0 \Rightarrow x = \frac{1}{2}, -1$$

$$\text{As } x > 0 \Rightarrow x = \frac{1}{2}$$

$$6 + \log_{2^{-1}} 2^{-1} = 6 + \frac{1}{2} \log_2 2 = \frac{6}{1} + \frac{1}{2} = \frac{13}{2}$$

37. (b) $f(x) = \log(x + \sqrt{x^2 + 1})$

$$f(-x) = \log[(-x) + \sqrt{(-x)^2 + 1}]$$

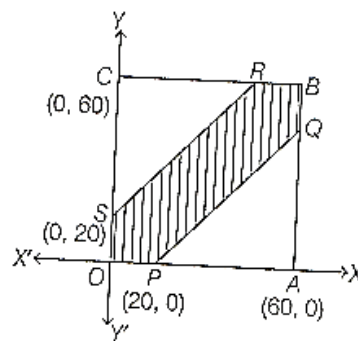
$$f(-x) = \log[\sqrt{x^2 + 1} - x]$$

$$= \log \left[\frac{(\sqrt{x^2 + 1} - x)(\sqrt{x^2 + 1} + x)}{\sqrt{x^2 + 1} + x} \right]$$

$$= \log \left(\frac{1}{x + \sqrt{x^2 + 1}} \right) = \log(x + \sqrt{x^2 + 1})^{-1}$$

$$= -\log(x + \sqrt{x^2 + 1}) = -f(x) \text{ odd function}$$

- 38. (a)** Let A and B arrive at the place of their meeting x minutes and y minutes after 6pm.



Their meeting is possible only if $|x - y| \leq 20$

$OABC$ is a square, where

$$A = (60, 0), C = (0, 60)$$

Considering the equality $|x - y| = 20$

The area representing the favourable cases

$$= \text{Area } OPQBRSO$$

$$= \text{Area of square } OABC - \text{Area of } \triangle PAQ$$

$$\text{Area of } \triangle SRC$$

$$= (60)(60) - (1/2)(40)(40) - (1/2)(40)(40)$$

$$= 3600 - 1600 = 2000 \text{ sq unit}$$

$$\text{Total way} = \text{Area of square } OABC$$

$$= (60)(60)$$

$$= 3,600 \text{ sq. unit}$$

$$\text{Required probability} = \frac{2000}{3600} = \frac{5}{9}$$

- 39. (a)** Total number of ways of choosing 3 numbers from the given set of numbers 1, 2, ..., 99 = ${}^{99}C_3$.

Let us divide the given numbers into 3 group G_1, G_2, G_3 as follows

$$G_1: 3, 6, 9, \dots, 99$$

$$[33 \text{ elements}]$$

$$G_2: 1, 4, 7, \dots, 97$$

$$[33 \text{ elements}]$$

$$G_3: 2, 5, 8, \dots, 98$$

$$[33 \text{ elements}]$$

Given that $a^3 + b^3 + c^3 - 3abc$ is to be divisible by 3.

If $a^3 + b^3 + c^3 - 3abc$ is divisible by 3, then which is possible in the following cases.

(ii) All the number belong to the second group.

(iii) All the number belong to the third group.

(iv) One number belong to the first group, one belong to second group and one belong to the third group.

$$\text{So, favourable number of cases are } 3 \cdot {}^{33}C_3 + ({}^{33}C_1)^3$$

$$\text{Required probability} = \frac{3 \cdot {}^{33}C_3 + ({}^{33}C_1)^3}{{}^{99}C_3}$$

- 40. (b)** There are A and B have to select as number from set of numbers $\{1, 2, \dots, 25\}$.

According to given condition, they will win prize if they choose the same numbers, i.e., then probability of choosing



same numbers is $1/25$.

Hence, the probability that they will not win a prize in a single trail

$$= 1 - \frac{1}{25} = \frac{25-1}{25} = \frac{24}{25}$$

41. (a) $\sin 18^\circ = \frac{\sqrt{5}-1}{4}$ and $\cos 36^\circ = \frac{\sqrt{5}+1}{4}$

$$\Rightarrow \sin^2 18^\circ = \left(\frac{\sqrt{5}-1}{4}\right)^2 \text{ and } \cos^2 36^\circ = \left(\frac{\sqrt{5}+1}{4}\right)^2$$

$$\alpha + \beta = \sin^2 18^\circ + \cos^2 36^\circ$$

$$= \frac{5+1-2\sqrt{5}}{16} + \frac{5+1+2\sqrt{5}}{16}$$

$$\alpha + \beta = \frac{6-2\sqrt{5}+6+2\sqrt{5}}{16} = \frac{12}{16}$$

$$\alpha\beta = \sin^2 18^\circ \cdot \cos^2 36^\circ$$

$$= \left(\frac{\sqrt{5}-1}{4}\right)^2 \cdot \left(\frac{\sqrt{5}+1}{4}\right)^2 = \left[\left(\frac{\sqrt{5}-1}{4}\right)\left(\frac{\sqrt{5}+1}{4}\right)\right]^2$$

$$\alpha\beta = \left(\frac{5-1}{16}\right)^2 = \left(\frac{4}{16}\right)^2 = \left(\frac{1}{4}\right)^2 = \frac{1}{16}$$

$$\Rightarrow x^2 - (\alpha + \beta)x + \alpha\beta = 0 \Rightarrow x^2 - \frac{12}{16}x + \frac{1}{16} = 0$$

$$\Rightarrow 16x^2 - 12x + 1 = 0$$

42. (a) Let the first term of GP be a and the common ratio be r

$$S_\infty = 2(a + ar)$$

$$\frac{a}{1-r} = 2a(1+r)$$

$$(1+r)(1-r) = \frac{1}{2}$$

$$1-r^2 = \frac{1}{2}$$

$$r^2 = \frac{1}{2}$$

$$r = \pm \frac{1}{\sqrt{2}}$$

43. (b) In question it should be given that these terms are in H.P.

Given that m^{th} term of HP

$$\frac{1}{n} = \frac{1}{a} + (m-1)d \quad \dots\dots(i)$$

And

$$n^{\text{th}} \text{ term } \frac{1}{m} = \frac{1}{a} + (n-1)d \quad \dots\dots(ii)$$

Subtract Eq. (ii) from Eq. (i), we get

$$\frac{1}{n} - \frac{1}{m} = d(m-n)$$

$$\text{Hence, } d = \frac{1}{mn}$$

Put this value in Eq. (i), we get $a = mn$

$$T_{m+n} = \frac{1}{\frac{1}{mn} + (m+n-1)\frac{1}{mn}} = \frac{mn}{m+n}$$

44. (b) If A is a skew symmetric matrix of odd order, then inverse does not exist.

$\Rightarrow$ If A is a skew symmetric matrix of even order, then A^{-1} is also a skew symmetric matrix.

$$A^T = -A$$

$$(A^T)^{-1} = (-A)^{-1}$$

$$(A^{-1})^T = -A^{-1}$$

45. (c) Given that mean of the squares of first n natural numbers be 1.

$$\Rightarrow \frac{n(n+1)(2n+1)}{6n} = 11 \Rightarrow (n+1)(2n+1) = 66$$

$$\Rightarrow 2n^2 + 3n + 1 = 66 \Rightarrow 2n^2 + 3n - 65 = 0$$

$$\Rightarrow n = \frac{-3 \pm \sqrt{9+520}}{4} \Rightarrow n = \frac{-3 \pm \sqrt{529}}{4}$$

$$\Rightarrow n = \frac{-3+23}{4} = \frac{20}{4} = 5 \text{ [Taking the +ve sign]}$$

$$\text{As, } n = \frac{-3-23}{4} = \frac{-26}{4} [n \text{ cannot be negative}]$$

$$\therefore n = 5$$

46. (b) $f(x) = \frac{x}{1+|x|} = \begin{cases} \frac{x}{1-x}, & x < 0 \\ \frac{x}{1+x}, & x \geq 0 \end{cases}$

$$\text{LHD} = \lim_{x \rightarrow 0^-} \frac{f(x)-f(0)}{x-0}$$

$$= \lim_{x \rightarrow 0^-} \frac{x}{1-x}$$

$$= \lim_{x \rightarrow 0^-} \frac{1}{1-x} = 1$$

$$\text{RHD} = \lim_{x \rightarrow 0^+} \frac{f(x)-f(0)}{x-0}$$

$$= \lim_{x \rightarrow 0^+} \frac{x}{1+x}$$

$$= \lim_{x \rightarrow 0^+} \frac{1}{1+x} = 1$$

Hence, LHD = RHD

So, it is differentiable everywhere.

47. (b) $I = \int_0^\pi x f(\sin x) dx$

$$I = \int_0^\pi (\pi - x) f(\sin(\pi - x)) dx$$

$$I = \int_0^\pi (\pi - x) f(\sin x) dx$$

$$I = \pi \int_0^\pi f(\sin x) dx - \int_0^\pi x f(\sin x) dx$$

$$I = \pi \int_0^\pi f(\sin x) dx - I$$

$$2I = \pi \int_0^\pi f(\sin x) dx$$

$$I = \pi/2 \int_0^\pi f(\sin x) dx$$

48. (c) Let $f: R \rightarrow R$ be defined by $f(x) = \begin{cases} x+2, & \text{if } x < 0 \\ |x-2|, & \text{if } x \geq 0 \end{cases}$

$$\int_{-2}^0 (x+2) dx + \int_0^2 (2-x) dx + \int_2^3 (x-2) dx$$

$$= \left[\frac{x^2}{2}\right]_{-2}^0 + 2[x]_{-2}^0 + 2[x]_0^2 - \left[\frac{x^2}{2}\right]_0^2 + \left[\frac{x^2}{2}\right]_2^3 - 2[x]_2^3 = 4.5$$

49. (b) $6x^2 - 2xy - 2y^2 = 0$

$$m_1 + m_2 = \frac{-2h}{b}, m_1 m_2 = \frac{a}{b}$$

$$m_1 + m_2 = \frac{2 \times 1}{-2}, m_1 m_2 = \frac{6}{-2}$$

$$m_1 + m_2 = -1, m_1 m_2 = -3$$

$$(m_1 - m_2)^2 = (m_1 + m_2)^2 - 4m_1 m_2$$

$$(m_1 - m_2)^2 = (-1)^2 - 4 \times (-3) = 1 + 12$$

$$m_1 - m_2 = \sqrt{13} \approx \frac{7}{2}$$

50. (d) Radius of circle change at $\frac{-2}{\pi}$ m/s

$$\frac{dA}{dt} = ? \text{ of circle } r = 10 \Rightarrow \frac{dr}{dt} = \frac{-2}{\pi} \text{ m/s}$$

$$\text{Area of circle (A)} = \pi r^2$$

$$\Rightarrow \frac{dA}{dt} = \pi \times 2r \frac{dr}{dt}$$

$$\Rightarrow \frac{dA}{dt} = \pi \times 2 \times 10 \times \frac{-2}{\pi} = -40 \text{ m}^2/\text{sec}$$

$$\Rightarrow \frac{dA}{dt} = -40 \text{ m}^2/\text{sec}$$

REASONING

51. (a) Artificial honey can be prepared as the word natural is mentioned in the advertisement. No comparison regarding the price of natural honey is given.

Also, no information regarding other companies is given. Hence, only assumption I is implicit.

52. (d) From the given statement it is clear the some talented Indian Scientist have migrated to America and also some Indian Scientists are talented.

Hence, conclusion (2) and (4) are true.



53. (b) Clearly, the paragraph best supports the statement (b).

54. (a) A runs $1\frac{2}{3}$ times as fast as B.

Hence, ratio of speed of A and B = 5:3 or we can say that A moves 5 m in the same time in which B moves 3 m.

Hence, A gains 2 m over B by running 5 m.

∴ Required distance = $\frac{5}{2} \times 80 = 200$ m

55. (a) Let the efficiency of one man be 'M' and of one boy be 'B', According to the question,

$$(2M+3B)10 = (3M+2B)8$$

$$\Rightarrow 20M+30B = 24M+16B$$

$$\Rightarrow 14B = 4M$$

$$\text{or } 2M = 7B$$

or We can say 10 boys can complete the work in 10 days.

Now, time taken to complete the work by 2 men and 1 boy = time taken to complete the work by 8 boys.

Now, 10 boys can do work in 10 days

∴ 8 boys can do it in $\frac{10 \times 10}{8} = 12.5$ days

56. (a) LCM of 2,3,4,5 and 6 = 60

Hence, Number between 1 and 1000 which are divisible by

60 = 16

as $1000 = 60 \times 16 + 40$

57. (c) Number of balls in 1st layer = 1

Number of balls in 2nd layer = 3 = 1 + 2

Number of balls in 3rd layer = 6 = 1 + 2 + 3

Number of balls in 4th layer = 10 = 1 + 2 + 3 + 4

Now, the n th term = $\frac{n(n+1)}{2} = \frac{n^2}{2} + \frac{n}{2}$

Sum of n terms of series = $\frac{1}{2}[\Sigma n^2 + \Sigma n]$

$$= \frac{1}{2} \left[\frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} \right] = \frac{n(n+1)(n+2)}{6}$$

$$\text{Now, } \frac{n(n+1)(n+2)}{6} = 8436$$

$$\text{or } n(n+1)(n+2) = 50616$$

The above condition is satisfied only for $n = 36$

58. (d) Speed while travelling from A to B = 50 km/h

Speed while travelling from B to A = 100 km/h

∴ Required average speed = $\frac{2 \times 50 \times 100}{50+100} \times \frac{1000}{3600} = 18.5$ m/s

Now, average velocity is 0 km/h as displacement is zero.

59. (d) Here,

$$D \xrightarrow{+1} E \xrightarrow{+3} H \xrightarrow{-1} G$$

$$R \xrightarrow{+1} S \xrightarrow{+3} V \xrightarrow{-1} U$$

$$J \xrightarrow{+1} K \xrightarrow{+3} N \xrightarrow{-1} M$$

$$L \xrightarrow{+1} M \xrightarrow{+4} Q \xrightarrow{-1} P$$

Hence, L M Q P is the odd one.

60. (d) Here, the pattern of the series is

15, 16, 34, 105, 424, 2124, 12756

$15 \times 1 + 1 = 16$

$$16 \times 2 + 2 = 34$$

$$34 \times 3 + 3 = 105$$

$$105 \times 4 + 4 = 424$$

$$424 \times 5 + 5 = 2125$$

Hence, 2124 is the wrong number.

61. (c) Age of man's sister = 12yr

∴ Age of man = 12 + 3 = 15yr

∴ Age of man's brother = 15 + 5 = 20yr

Hence, age of man's mother = 20 × 2 = 40yr

62. (b) The pattern of the series is,

$$(1)^2 = 1 \times 3 = 3$$

↓ +2

$$(2)^2 = 4 \times 5 = 20$$

↓ +2

$$(3)^2 = 9 \times 7 = 63$$

↓ +2

$$(4)^2 = 16 \times 9 = 144$$

↓ +2

$$(5)^2 = 25 \times 11 = 275$$

↓ +2

$$(6)^2 = 36 \times 13 = \boxed{468}$$

63. (b) Two members (T and Z) are serving committee B.
∴ Three members would serve committee A. and hence, 4 members would serve on committee C.

64. (a) T, S and X serve committee B.

∴ 4 members would serve committee A.

x. Now, Z, Y and X cannot serve committee A.

Hence, Z and Y serve on committee C.

65. (d) R is the only individual serving committee B.

∴ 2 members would serve committee A and the rest would serve committee C.

Now, T, S and R cannot serve on committee C.

Hence, T and S serve on committee A.

66. (d) Let the total population be 100.

∴ Number of adults = $\frac{2}{5} \times 100 = 40$

and number of children = 60

Now, 60% of adults are literate.

75×5

∴ Number of literate adults = $\frac{60}{100} \times 40 = 24$

and number of literate children = $\frac{85}{100} \times 60 = 51$

∴ Total number of literates = 24 + 51 = 75

∴ Required percentage = $\frac{75}{100} \times 100 = 75\%$

67. (d) Total number of students = 50

Number of students who speak English = 21 and number of students who speak both English and Hindi = 10

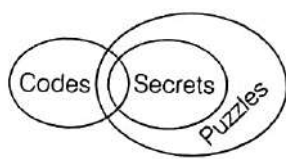
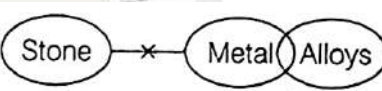
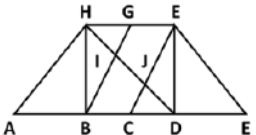
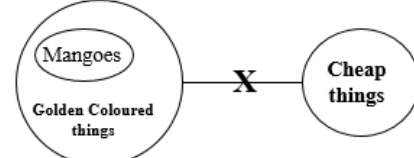
∴ Number of students who speak only English = 21 - 10 = 11

and number of students who speak Hindi = 50 - 11 = 39

and number of students who speak only Hindi = 39 - 10 = 29



68. (a) The last digit can be obtained by multiplying the last digits of the given numbers.
Here, 82×85 , i.e. $2 \times 5 = 10$
Hence, the last digit would be 0.
69. (a) As there is a considerable decrease in hiring of engineering graduates, a large number of engineering seats remains vacant.
70. (d) At 3 O' clock both the hands of the clock are 15 min space apart.
To coincide, the minute hand must gain 15 min over the hour hand.
Now, 55 min are gained in 60 min.
 $\therefore$ 15 min would be gained in $\frac{60}{55} \times 15 = 16\frac{4}{11}$
Hence, the needles will coincide after $16\frac{4}{11}$ min.
71. (c) As, $\log_3 2, \log_3(2^x - 5), \log_3(2^x - \frac{7}{2})$ are in arithmetic progression.
 $\therefore \log_3(2^x - 5) - \log_3 2 = \log_3(2^x - \frac{7}{2}) - \log_3(2^x - 5)$
 $= \log_3(\frac{2^x - 5}{2}) = \log_3(\frac{2^x - 7}{2})$
 $[\because \log(\frac{a}{b}) = \log a - \log b]$
 $\Rightarrow (2^x - 5)(2^x - 5) = (2^x - \frac{7}{2}) \times 2$
 $\Rightarrow (2^x - 5)^2 = 2^{x+1} - 7$
 $\Rightarrow 2^{2x} + 25 - 2 \times 2^x \times 5 = 2^{x+1} - 7$
 $\Rightarrow 2^{2x} + 2^{x+1} \times 5 - 2^{2x} = 32$
 $\Rightarrow 2^{x+1}(6) - 2^{2x} = 32$
 $\Rightarrow 2^x(12 - 2^x) = 32$
The above equation is satisfied for $x = 3$
72. (c) For a number to be divisible by 4 the last two digits must be divisible by 4.
 $\therefore$ The possible cases are
- | | | |
|---------|--|---------|
| ---- 12 | | ---- 16 |
| ---- 24 | | ---- 32 |
| ---- 36 | | ---- 52 |
| ---- 56 | | ---- 64 |
- $\therefore$ Value of $n = 4 \times 3 \times 2 \times 1 \times 8 = 192$
73. (d) First number divisible by 3 between 100 and 200 = 102
Last number divisible by 3 between 100 and 200 = 198
Total numbers divisible by 3 between 100 and 200 = 33
Out of which 105, 126, 147, 168 and 189 are divisible by 7 and from remaining numbers 102, 108, 114, 120, 132, 138, 144, 150, 156, 162, 174, 180, 186, 192 and 198 are even numbers.
 $\therefore x = 33 - 20 = 13$
74. (b) Pipe A can fill a cistern in 37.5 min
 $\therefore$ Part of cistern filled by pipe A in 1 min = $\frac{1}{37.5}$
Similarly, part of cistern filled by pipe B in 1 min = $\frac{1}{45}$
Now, the cistern will be filled in half an hour if pipe B is turned off after some time.
Let pipe B is turned off after min.
 $\frac{30}{37.5} + \frac{x}{45} = 1$
 $\therefore 0.8 + \frac{x}{45} = 1$
 $x = 45 \times 0.2 = 9$ min

75. (b) According to the given information,
- | Person | Shirt's Colour | Pant's Colour |
|--------|----------------|---------------|
| A | Orange | Yellow |
| B | Black | Orange |
| C | Blue | Green |
- Clearly, C wore pant and shirt of Green and Blue colour, respectively.
76. (b) In each column the element on the left hand side remains the same while the element on the right rotates 90° clock wise in each step.
77. (b) T-shirt and shoes both items are worn on the body. Similarly, rack and sofa are kept inside the house.
78. (d)
- 
- Conclusion**
I. ($\checkmark$) II. ($\checkmark$)
Hence, both conclusions I and II are true.
79. (b)
- 
- Conclusion**
I. ($\times$) II. ($\checkmark$)
Hence, only conclusion II is true.
80. (c)
- 
- Total number of triangles = 14
($\triangle ABH, \triangle HIB, \triangle HGI, \triangle FJD, \triangle CJD, \triangle FDE, \triangle HFI, \triangle ADIB, \triangle HFD, \triangle DBH, \triangle AHD, \triangle CFE, \triangle CFD$ and $\triangle BHG$)
81. (b)
- 
- Conclusion**
I. ($\times$) II. ($\checkmark$)
Hence, only conclusion II is true.
82. (d) All the statements are necessary to find the profit percentage. Let the marked price of the item be ₹ x.
From statement III,
Cost price of article = ₹5000
From statement II,
If discount is not given, profit percentage = 20%

94. (b)



$$\begin{aligned}
 2(16)^2 + 3(16) - 560 &= 0 \\
 \Rightarrow 512 + 48 - 560 &= 0 \\
 \Rightarrow 560 - 560 &= 0 \\
 \Rightarrow (565)_{10} &= (1065)_{R_2} \\
 \Rightarrow R_2^3 + 6R_2 + 5 &= 565 \\
 \Rightarrow R_2^3 + 6R_2 - 560 &= 0 \\
 \text{If we put } R_2 &= 8, \text{ then} \\
 \Rightarrow (8)^3 + 6(8) - 560 &= 0 \\
 \Rightarrow 512 + 48 - 560 &= 0 \\
 \Rightarrow 560 - 560 &= 0.
 \end{aligned}$$

95. (b)

In NOR gate, if one input is 0 and other is 1. Then, the output will be 0. If both inputs are low. Then, the output is 1. So, NOR gate is not a answer.

In NAND gate, if one input is 0 and other is either 0 or 1. The output is high. So, NAND gate is a answer.

96. (c) $\sqrt{(224)r} = (13)r$

On squaring both sides,

$$(224)r = (13)^2 r$$

Now, converting to decimal number

$$2r^2 + 2r + 4 = (r + 3)^2$$

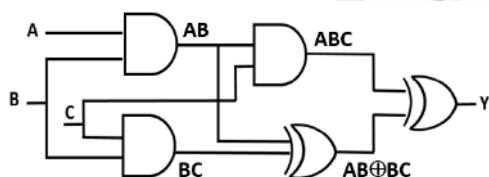
$$2r^2 + 2r + 4 = r^2 + 9 + 6r$$

$$r^2 - 4r - 5 = 0$$

Root of the above equation is 5, -1

Therefore, radix = 5

97. (c)



$$\text{So, } Y = ABC \oplus (AB \oplus BC)$$

$$Y = B(C + A)$$

98. (d)

The exclusive NOR and exclusive OR of x and y is given by the expressions.

$$x \circ y = xy + x'y' \text{ (exclusive NOR or XNOR)}$$

$$x \oplus y = x'y + xy' \text{ (exclusive OR or XOR)}$$

So, option (A) is automatically true.

Option (B) is $x \oplus y'$ which evaluates to $x'y' + (x)(y)'$

$$= x'y' + xy \text{ which is equivalent to } x \circ y$$

Option (C) is $x' \oplus y$ which evaluates to

$$(x')y + x'y' = xy + x'y' \text{ which is equivalent to } x \circ y$$

Option (D) is $x' \oplus y'$ which evaluates to $(x')y' + x'(y)'$

$$= xy' + x'y \text{ which is not equivalent to } x \circ y.$$

99. (c)

Logical error.

100. (b)

$$(43)_x = (y3)_8$$

Since, a number in base K can only have digits from 0 to (K - 1) we can conclude that

$$x \geq 5 \text{ and } y \leq 7$$

Now, the original equation, when converted to decimal base gives:

$$4x^1 + 3x^0 = y(8^1) + 3(8^0)$$

$$\Rightarrow 4x + 3 = 8y + 3$$

$$\Rightarrow x = 2y$$

So, we have the following constraints

$$x \geq 5$$

$$y \leq 7$$

$x = 2y$, x and y are integers.

The set of values of (x, y) that satisfy these constraints are:

(6, 3), (8, 4), (10, 5), (12, 6), (14, 7)

So, the number of possible solution is 5.

ENGLISH

101.	A	102.	B	103.	D	104.	D	105.	D
106.	A	107.	B	108.	D	109.	C	110.	B
111.	C	112.	B	113.	C	114.	C	115.	D
116.	D	117.	B	118.	A	119.	A	120.	C



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